

Introduction to Excel for Data Analytics

Welcome to the first module of our Data Analytics journey. Before we dive into advanced tools and techniques, we start with Excel since it is one of the most widely used tools in the world for analyzing data. From small startups to large enterprises, everyone uses Excel to store, organize, and gain insights from data.

How to Get Microsoft Excel

- Most laptops come with Microsoft Office pre-installed
 - If not, you can easily download Excel online 😊
 - Excel is available on Windows, macOS, and even on tablets and mobile devices
 - You can also use the free online version at **Office.com**
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Certificate of Completion

Once you successfully finish the entire course, you will receive an official **Certificate of Completion** that you can showcase on LinkedIn and your resume to demonstrate your data analytics skills.

Excel: Visual Walkthrough

Before doing any analytics in Excel, let's get familiar with the main interface and layout. This will help you navigate quickly and efficiently while working on datasets.

Starting Excel

When you open Excel, you will see two main options:

1. Blank Workbook

This is the most commonly used option. It opens a fresh sheet with rows and columns where you can directly start entering and analyzing data.

2. Templates

Excel also offers built-in templates such as:

- Budget planners
- Calendars
- To-Do lists
- Project trackers

Templates can save time when you need a ready-made structure.

✓ For now, choose "Blank Workbook" to get started

Excel Interface Overview

Once your workbook opens, take a moment to explore the interface:

The Ribbon and Tabs

At the top of Excel, you'll see a **ribbon** with multiple tabs, each containing different tools:

Tab Name	Purpose
Home	Basic tasks like formatting text, alignment, copy-paste, number formats
Insert	Add charts, tables, images, shapes
Draw <i>(may not be available on every version)</i>	Drawing tools for touch/stylus users
Page Layout, Formulas, Data, Review, View	More advanced functionality we will explore later

The ribbon is your toolbox for everything you want to do in Excel.

Hiding the Ribbon

If the ribbon feels distracting:

- **Double-click** on any tab name (Home/Insert etc.)
- The ribbon collapses and gives you more working space
- Double-click again to bring it back

This is useful when working on smaller screens.

Quick Access Toolbar

You'll also notice a small toolbar at the very top left of Excel. This is the **Quick Access Toolbar**.

You can customize it to include your frequently used actions like:

- Save

- Undo
- Redo
- Quick Print
- New Workbook

Right-click any command > **Add to Quick Access Toolbar** This helps speed up your workflow.

Workbooks and Sheets

Excel organizes data in a structured format using **workbooks**, **sheets**, and **cells**.

What is a Workbook?

- A **workbook** is an Excel file
- It can contain **one or multiple sheets**
- Example: A project file with separate sheets for Sales, Expenses, and Summary

You can think of a workbook like a notebook with multiple pages.

What is a Sheet?

- A **sheet** (or **worksheet**) is a single page inside a workbook
- It is made up of **rows** and **columns**
- Each sheet has a separate space to store different sets of data

By default, most new workbooks start with **Sheet1**.

What is a Cell?

- A **cell** is the smallest unit of Excel where data is stored
- It is identified by a **column letter + row number**
 - Example: `A1` , `B15` , `C3`

Cells can store:

- Text
 - Numbers
 - Dates
 - Formulas
 - Functions
-

Naming Workbooks and Sheets

Rename Workbook

Rename your file when saving:

- Go to **File > Save As**
- Enter a meaningful name like: **Sales_Report_2025.xlsx**

Rename a Sheet

- **Double-click** the sheet name at the bottom
- Type a new name Example: Sales, Employees, Products, etc.

Always use names that clearly describe the data inside.

Inserting Data into a Sheet

To insert data:

1. Click any cell
2. Type the value
3. Press **Enter** or **Tab**

Excel automatically adjusts cell size based on data. You can also adjust column width and row height manually later.

Data Size and Performance

- Excel is extremely powerful but **can lag with very large datasets**
 - Especially when using many formulas or graphics
 - For big data analytics, tools like **SQL, Python, and Power BI** work faster (which we will cover later in this course)
-

Saving Workbooks and AutoSave

- Frequently press `Ctrl + S` to avoid losing work
 - If using **OneDrive or SharePoint**, AutoSave can automatically update changes
 - Local files need manual saving
-

Excel Size Limits

Excel supports a huge amount of data:

- **Rows:** 1,048,576
- **Columns:** 16,384 (Column XFD)

That means each sheet can handle millions of cells of data.

Saving and Opening Workbooks

Managing files is an essential part of working with Excel. You should know how to create, save, open, and share your Excel files properly.

Save vs Save As

Save

- Updates the currently open workbook
- If the file is already saved once, all changes are stored in the same file

Shortcut: `Ctrl + S`

Save As

- Used for **new** workbooks that have never been saved before
 - Also useful when saving a **copy** with a different name or location **Note:** For a new document, clicking **Save** will automatically open **Save As** first
-

Creating New Files

- Go to **File > New > Blank Workbook**
 - Or simply use the shortcut: `Ctrl + N`
-

Opening Workbooks

You can open existing Excel files using:

- **File > Open > Browse** to select a file
 - **Recent files** list to quickly open previously used files **Shortcut:** `Ctrl + O`
-

Distributing (Sharing) Files

After saving your workbook, you can share the `.xlsx` file through:

- Email
 - Cloud services like OneDrive, Google Drive
 - USB drives
 - Teams or other collaboration tools
-

Difference between `.xlsx` vs `.xls` File Types

Format	Used Since	Notes
<code>.xlsx</code>	2006 onwards	Default format in modern Excel, supports more features
<code>.xls</code>	Before 2006	Older format, limited compatibility

Excel supports many file types, but in this course, we will mainly use `.xlsx`.

Closing Workbooks

When done with a file:

- Go to **File > Close**
- Or use the shortcut: `Ctrl + W`

(This closes only the workbook, not Excel itself.)

Shortcut Summary

Action	Shortcut
Save	<code>Ctrl + S</code>
Open	<code>Ctrl + O</code>
New Workbook	<code>Ctrl + N</code>
Close Workbook	<code>Ctrl + W</code>

Adding Data With Headers in Excel

Entering data in Excel is simple and efficient. Each cell can store numbers, text, dates, times and more.

Typing Data into Cells

1. Click on a cell to select it.
2. Start typing your value.
3. Press Enter to confirm the entry.

Navigating While Entering Data

- Press **Tab** to move to the next cell in the same row.
- Press **Enter** to move to the next row in the same column.
- Use the arrow keys for quick navigation across the sheet.

Inserting Rows and Columns

You can add new rows and columns when you need more space for data.

- Right click on a row number and select **Insert** to add a row above it.
- Right click on a column letter and select **Insert** to add a column to the left.

When working with tables, always include **headers** for the columns. Headers make sorting, filtering and data analysis much easier.

Quick Access Toolbar Shortcut

Any command that you frequently use can be added to the Quick Access Toolbar.

- Right click on a command in the Ribbon.

- Select **Add to Quick Access Toolbar**.

This toolbar helps keep your most used commands accessible at all times.

Alignment, Colors and Fonts in Excel

Alignment

- **Left Align** keeps text and data toward the left side of the cell. Text is left aligned by default.
- **Right Align** positions values to the right. Numbers are usually right aligned.
- **Center Align** places content in the middle of the cell.

You can either align text horizontally (left, center, right) or vertically (top, middle, bottom) within a cell.

Colors

- You can change **font color** to highlight important text.
- You can change **fill color** to shade the background of cells for better visibility.

Fonts

- You can choose different **font styles, sizes** and **bold, italic or underline** to format text clearly.

Format Painter

- **Format Painter** copies formatting from one cell to another.
- Select a formatted cell, click Format Painter, then apply it to other cells to keep styles consistent.

Understanding Excel Cell Formats

Excel stores data in cells but controls how that data is displayed using **formats**. Formatting does not change the actual value, only how it looks.

Common Number Formats

- **General** Default format. Excel decides how to show the value.
- **Number** Can show decimals and comma separators. Example: 1,500.00
- **Currency** Shows currency symbols. Example: ₹1,500.00
- **Accounting** Aligns currency symbols and decimal points neatly in a column.
- **Percentage** Converts a number to percent form. Example: 0.25 becomes 25 percent
- **Fraction** Shows numbers as fractions. Example: 0.5 becomes 1/2
- **Scientific** Displays very large or small numbers using E notation. Example: 2.5E+06

Date and Time Formats

Dates and times are stored as numbers in Excel but displayed in readable formats.

- Dates: 04 Dec 2025, 4/12/25
- Time: 12:30 PM, 23:30

Text Format

Content is treated as plain text. Numbers stored as text do not calculate.

Choosing Formats

Select the cell or range Home tab > Number group > Choose the desired format

Performing Calculations in Excel

Excel is often called calculator on steroids because it can perform calculations instantly on large amounts of data.

Entering Formulas

All formulas start with an = sign.

- Example using numbers: `=10+20`
- Example using cell references: `=A1+B1` When you double click the formula cell, Excel highlights the referenced cells in different colors so you can check the formula easily.

AutoSum and Basic Functions

Excel provides built in functions for quick calculations:

- **SUM** adds numbers
- **AVERAGE** calculates the mean value
- **AutoSum** button automatically selects a range and inserts a SUM formula

Fill Handle for Quick Calculations

The small square at the bottom right of a selected cell is called the **Fill Handle**. You can drag it to copy a formula or pattern:

- If you select 12 then drag, it may create a sequence like 13, 14, 15
- If you select 1 and drag, it may repeat like 1, 1, 1, 1 depending on the pattern Excel recognizes

Copying Formulas

Excel uses **relative references** by default. When you copy a formula down, references change automatically based on position.

- Example: `=A1+B1` becomes `=A2+B2` in the next row

Use **absolute references** when you want a cell reference to stay fixed.

- Use `$` symbol Example: `=$A$1+B1` keeps A1 constant when copying

Relative and Absolute References in Excel

Quick Tip: You can create a new Excel sheet in Windows by right clicking on the desktop or inside a folder, selecting **New**, then **Microsoft Excel Worksheet**.

What Are Relative References

Relative references change automatically when you copy a formula to another cell.

Example: `=A1+B1` becomes `=A2+B2` when copied one row down.

What Are Absolute References

Absolute references stay fixed even when you copy the formula. Example:

`=$A$1+B1` keeps A1 constant everywhere.

The Dollar Symbol

The `$` sign is used to lock a reference.

- `$A$1` locks both column A and row 1
- `$A1` locks the column only
- `A$1` locks the row only

What Happens When You Copy

- **Relative reference copied:** Excel shifts the reference based on the new location. Formulas update automatically.
- **Absolute reference copied:** Excel keeps the locked part unchanged, so the formula always points to the same cell.

This makes it easy to create formulas that adjust when needed or stay fixed when required.

Order of Operations in Excel

Excel follows a standard order of operations when solving formulas. This is known as **BODMAS** or **PEMDAS**.

- **B / P**: Brackets or Parentheses
- **O / E**: Orders or Exponents
- **D**: Division
- **M**: Multiplication
- **A**: Addition
- **S**: Subtraction

Excel uses this sequence to decide which part of a formula to calculate first.

Use Parentheses to Be Sure

Always use parentheses to control the calculation and avoid mistakes. Example:

`=(10+5)*2` ensures the addition happens before multiplication.

Evaluate Formula Tool

Excel provides an **Evaluate Formula** option to check how Excel solves a formula step by step. You can find it under: Formulas tab > Formula Auditing group > Evaluate Formula

This helps you understand and debug complex formulas.

Functions in Excel

Functions are built in formulas that perform specific calculations. Instead of writing long formulas yourself, you use a function that already knows how to do the work.

Structure of a Function

Every function follows the same pattern:

```
=FUNCTION_NAME(arguments)
```

Example:

```
=SUM(A1:A10)
```

Using Entire Rows and Columns

You can apply functions to full rows and columns:

- Sum an entire column B `=SUM(B:B)`
- Sum an entire row 1 `=SUM(1:1)`

Multiple Arguments

Functions can take more than one argument:

```
=SUM(J20:J22,4)
```

This adds the range J20 to J22 and the number 4.

Function Arguments Window

Clicking the **F(x)** button next to the formula bar opens the Function Arguments window. This helps you enter values step by step.

You can also click **Help on this function** inside that window to open detailed documentation.

Autocomplete Suggestions

When you start typing a function name, Excel shows suggestions. This makes it easy to find the correct function and reduces spelling mistakes.

Where to Find Functions

Go to: Formulas tab > Function Library group > Insert Function

This is where you can browse functions by category such as Math, Text, Logical and more.

Commonly Used Excel Functions

Excel provides many built in functions, but a few are used in almost every worksheet.

SUM Function

The SUM function adds numbers quickly.

```
=SUM(A1:A10)
```

You do not need to type each cell reference again and again. Use **AutoSum** to add a range instantly. Shortcut: **Alt + =** inserts a SUM formula automatically.

MIN Function

Finds the smallest value in a range.

```
=MIN(A1:A10)
```

You can also use an entire column:

```
=MIN(B:B)
```

MAX Function

Finds the largest value in a range.

```
=MAX(A1:A10)
```

AVERAGE Function

Calculates the mean of numbers.

```
=AVERAGE (A1 : A10)
```

COUNT Function

Counts how many cells contain numbers.

```
=COUNT (A1 : A10)
```

Copying Formulas with Auto Fill Handle

Use the fill handle (small square at the bottom right of a selected cell) to drag and copy formulas across rows or columns. Excel updates the references automatically.

Modifying an Excel Sheet

Moving, Copying and Deleting Cells

- You can **drag and drop** a cell to move its contents.
- You can **drag and drop** a **selected range** to move multiple cells together.
- Use **Cut (Ctrl + X)** and paste to move data. Press **Ctrl + D** to deselect after pasting.
- Deleting cells removes the data and may shift surrounding cells.

Inserting Rows and Columns

- Use **Ctrl + Plus (+)** to insert a row or column.
- Use **Ctrl + Minus (-)** to delete a row or column.
- If you do not have a number pad, use **Ctrl + Shift + =** to insert.

Resizing Rows and Columns

- Resize rows or columns by dragging their borders.
- If a column is too narrow, Excel may show #####. Increase the column width to display the value correctly.

Hide and Unhide

- You can hide rows or columns to reduce clutter.
- Hidden rows or columns can be unhidden later from the same menu.

Working with Worksheets

- Deleted worksheets cannot be undone.

- Hold **Ctrl** and drag a worksheet tab to create a copy.
- You can move a worksheet to a different workbook by dragging its tab to the other workbook window.

Formatting Data in an Excel Sheet

Font Formatting

- You can apply **bold**, **italic** and **underline** to text.
- Change **font color** to highlight important values.
- Use **cell fill color** to shade the background of cells.

Themes

- Go to **Page Layout > Themes** to apply a consistent look across the entire worksheet.
- Themes control fonts, colors and effects together.

Borders

- Borders help separate and organize data.
- You can change **border line color** and **line styles**.
- Use **More Borders** to access advanced border options.

Number Formatting

- Percentages can be applied directly or by using the **fraction** format when appropriate.
- Formatting affects display only, not the underlying value.

Cell Styles

- **Cell Styles** provide pre defined formatting for titles, headings, totals and more.
- Using styles helps keep formatting consistent across the sheet.

Conditional Formatting in Excel

Conditional formatting changes the appearance of cells based on their values. It helps you quickly spot patterns, trends and outliers.

Highlight Cells Rules

- Highlight cells based on conditions such as greater than, less than or equal to.
- You can also highlight duplicate values, text or dates.

Data Bars

- Data bars add visual bars inside cells.
- The bar length represents the value relative to other cells in the range.

Manage Rules

- Use **Manage Rules** to view all conditional formatting rules applied to a sheet.
- You can control rule priority and where each rule applies.

Edit Rules

- Edit a rule to change conditions, formatting or ranges.
- This is useful when your data or logic changes.

Clear Rules

- Clear rules to remove conditional formatting.
- You can clear rules from selected cells or from the entire worksheet.

Sorting and Filtering in Excel

Sorting and filtering help you organize and analyze data more effectively.

Sorting Data

- Go to **Data > Sort** to arrange data in ascending or descending order.
- You can sort by numbers, text, dates or custom criteria.

Filtering Data

- Go to **Data > Filter** to enable filters on column headers.
- Filters allow you to view only the data that meets specific conditions.

Keyboard Shortcut

- **Alt + D + F + F** is the shortcut to apply filters.
- This shortcut toggles the filter on and off.
- Once filters are applied, you can also use them to sort data quickly from the header dropdowns.

Generating Random Numbers in Excel

Excel provides several functions to generate random numbers for testing and analysis.

RAND Function

- `=RAND()`
- Returns a random decimal number between 0 and 1.
- The value changes every time the worksheet recalculates.

RANDBETWEEN Function

- `=RANDBETWEEN(min, max)`
- Returns a random whole number between the specified minimum and maximum values.
- Example: `=RANDBETWEEN(10, 50)`

RANDARRAY Function

- `=RANDARRAY(rows, columns, min, max, whole_number)`
- Generates an array of random numbers.
- Example: `=RANDARRAY(5, 1, 1, 100, TRUE)`

Random Number Using Min and Max Technique

You can generate a random number between any two values using:

```
=min + (RAND() * (max - min))
```

Example:

```
=10 + (RAND() * (50 - 10))
```

This returns a random decimal number between 10 and 50.

QuickTip: Selecting Data Quickly

- Use **Ctrl + Shift + Down Arrow** to select all continuous data below the current cell.

Images and Shapes in Excel

Excel allows you to add visual elements like images, shapes and diagrams to make your sheets more informative.

Inserting Images

- Go to **Insert > Illustrations > Pictures** to insert pictures.
- **Online Pictures** are powered by Bing Image Search.

Working with Shapes

- Go to **Insert > Shapes** to add shapes.
- Available shape categories include:
 - Basic Shapes
 - Block Arrows
 - Equation Shapes
 - Flowchart
 - Stars and Banners
 - Callouts

Resizing and Adjusting Shapes

- **White markers** are used to resize the shape.
- **Orange markers** are used to adjust the thickness or internal structure of certain shape elements.

Shape Effects

- Use **Shape Effects** to add shadows, reflections and other visual styles.

SmartArt

- Go to **Insert** > **SmartArt** to insert diagrams such as processes, hierarchies and relationships.
- SmartArt is useful for explaining workflows and structured information visually.

Charts in Excel

Excel charts help visualize data and identify trends quickly.

Common Chart Types

Column, Line, Pie, Bar, Area and Scatter charts are commonly used to compare values, show trends, display proportions and analyze relationships between data.

- Column Charts: Useful for comparing values across categories.
- Line Charts: Ideal for showing trends over time.
- Pie Charts: Show proportions of a whole.
- Bar Charts: Similar to column charts but oriented horizontally.
- Area Charts: Emphasize the magnitude of change over time.
- Scatter Charts: Show relationships between two variables.

Inserting a Chart

- Select the data.
- Go to **Insert > Charts** and choose the required chart type.

Chart Design and Data

- Use **Chart Design > Select Data** to modify the data source.
- You can add or remove data series from here.

Formatting Charts

- Use **Format Axis** to change scale, labels and number formatting.
- You can format chart elements such as colors, fonts, gridlines and legends.

Moving Charts

- Go to **Chart Design > Move Chart** to place the chart on a new worksheet or move it back into the current sheet.

Pie Chart Data Labels

- Use **Format Data Labels** to show values, percentages or category names.
- To pull out a slice, click the pie once, click the slice again and drag it outward.

Printing an Excel Document

Printing in Excel gives you control over layout, scaling and what content appears on paper.

Print Preview and Page Breaks

- After opening **Print Preview** and returning to the sheet, Excel displays **page breaks**.
- Page breaks show how the sheet will be split across printed pages.

Printing Charts

- If a chart is selected while printing, **only the chart** will be printed.

Scaling Options

- Use **custom scaling** to fit content on one page or a specific number of pages.
- Scaling options are available in the Print settings.

Page Setup and Margins

- Go to **Page Setup > Margins** to adjust spacing around the content.
- Proper margins help prevent content from being cut off.

Page Layout and Zoom

- Use **Page Layout** view to see how the sheet fits on pages.
- Go to **View > Zoom** to adjust how large or small the sheet appears on screen.

Headers and Footers

- You can add **page numbers** using Header and Footer options.
- This is useful for multi page printouts.

Print Selection

- Use **Print Selection** to print only the selected cells instead of the entire worksheet.

Microsoft Print to PDF

- Select **Microsoft Print to PDF** as the printer to save the document as a PDF file instead of printing on paper.

Templates in Excel

You can use templates in Excel to create pre designed spreadsheets for various purposes. Templates save time and help maintain consistency across documents.

Using Built In Templates

Excel provides ready made templates for budgets, invoices, trackers, calendars and more. - Go to **File > New** - Search for the required template type and select one

Creating a Template from an Existing Sheet

- Right click on the sheet tab
- Select **Move or Copy**
- Choose **New Book**
- Enable **Create a copy**

Saving as an Excel Template

- Go to **File > Save As**
- Select file type ****Excel Template (*.xltx)****
- Save the file

Custom Office Templates Location

Save your templates in the following directory on your computer (Windows):

```
C:\Users\YourUsername\Documents\Custom Office Templates
```

Using this location is important because when you create a new document in Excel and select **Personal** under the template options, Excel shows the templates saved in this folder.

Referencing Data Across Worksheets in Excel

Excel allows you to use data from one worksheet inside another worksheet within the same workbook.

Basic Cross Worksheet Reference

To reference a cell from another sheet, use:

```
=SheetName!CellReference
```

Example:

```
=Sheet2!A1
```

Using References in Formulas

You can perform calculations using cells from different sheets:

```
=Sheet1!A1 + Sheet2!A1
```

Referencing Ranges

You can also reference a full range from another worksheet:

```
=Sheet1!A1:A10
```

How Excel Helps You

- Type `=` and switch to another sheet.
- Click the required cell or range.
- Excel automatically inserts the correct reference.

Important Notes

- If the sheet name contains spaces, Excel adds quotes automatically:

```
= 'Sales Data' !B2
```

- Renaming a sheet updates references automatically.

Cross worksheet referencing is useful for summaries, dashboards and keeping raw data separate from calculations.